Methods of Point Estimation

Maximum Likelihood Estimation (MLE)

Maximum Likelihood Estimation: Introduction

- Maximum Likelihood Estimation (MLE) is a method for finding the **best guess** for unknown parameters of a model.
- Imagine you observe some data (e.g., heights of people, number of customers per day, etc.).
- You assume the data comes from a known type of distribution (like Normal, Poisson, etc.), but you don't know the parameters (such as mean or variance).
- **The method of maximum likelihood says:**

  "Choose the parameters that make the observed data most likely."
- In other words, find the parameter values that would make the data you saw most probable.

Maximum Likelihood Estimation (MLE)

- Introduced by Ronald Fisher in the 1920s.
- **Idea**: Choose the parameter value that maximizes the likelihood of observing the given data.
- **Likelihood Function:** Let $\boldsymbol{X} = [X_1, \dots, X_N]^T$ be a random vector drawn from a joint PDF $f(\mathbf{x}; \theta)$, and let $\boldsymbol{x} = [x_1, \dots, x_N]^T$ be the realizations. The likelihood function is a function of the parameter θ given the realizations $\boldsymbol{x}$:

$$L(\theta) = f(x_1, x_2, ..., x_n; \theta) = f(x_1, \theta) f(x_2, \theta) \cdots f(x_n, \theta) = \prod_{i=1}^{n} f(x_i; \theta)$$

Interpretation of Likelihood

- The likelihood is not a probability — it is a function of θ.
- It measures how "plausible" different parameter values are, given the observed data.
- Higher likelihood means the parameter better explains the observed data.

Properties of MLEs

- **Consistent:** MLEs converge to the true parameter value as the sample size increases.
- **Efficient:** MLEs achieve the lowest possible variance asymptotically (large sample sizes).
- **Sufficient:** MLEs often capture all the information about the parameter contained in the sample.
- **Not necessarily unbiased:** MLEs can be biased, especially in small samples.

Example:

- Estimating variance σ^2 of a normal distribution using:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

- This MLE is **biased** because it underestimates the true variance.
- (The unbiased estimator uses $\frac{1}{n-1}$ instead of $\frac{1}{n}$.)

Procedure for Finding an MLE

1. Write the likelihood function $L(\theta)$.
2. Take the natural logarithm to get the log-likelihood $\ell(\theta)$.
3. Differentiate $\ell(\theta)$ with respect to θ.
4. Solve $\frac{d\ell}{d\theta} = 0$ to find the estimator $\hat{\theta}$.
5. Verify that it maximizes the likelihood (second derivative test, etc.).

Examples on MLE

Example 1: Given n independent experiments, each following a geometric distribution, where the number of trials until the first success is observed in each experiment, find the Maximum Likelihood Estimator (MLE) of the parameter p (probability of success).

The probability mass function (PMF) of the geometric distribution is:

$$P(X = x) = (1 - p)^{x-1} p \quad \text{for} \quad x = 1, 2, 3, \ldots$$

where:

- p is the probability of success,
- x is the number of trials until the first success.

Examples on MLE

Solution: The likelihood function for n independent experiments, where $X_1, X_2, \ldots, X_n$ are the observed number of trials until success, is:

$$L(p) = \prod_{i=1}^{n} (1-p)^{x_i - 1} p = p^n \cdot (1-p)^{S-n}$$

where $S = \sum_{i=1}^{n} x_i$ is the sum of the observed trial numbers.

The log-likelihood function is:

$$\ell(p) = \log(L(p)) = n \log(p) + (S-n) \log(1-p)$$

Examples on MLE

To find the MLE of p, we take the derivative of the log-likelihood function with respect to p, set it to zero, and solve for p:

$$\frac{d}{dp}\ell(p) = \frac{n}{p} - \frac{S-n}{1-p} = 0$$

Solving for p:

$$\frac{n}{p} = \frac{S-n}{1-p}$$

Cross-multiplying and simplifying:

$$n(1-p) = (S-n)p \quad \Rightarrow \quad p = \frac{n}{S} = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\bar{x}}$$

The MLE of p is the reciprocal of the mean of the number of trials.

Examples on MLE

Example 2: Suppose $X_1, X_2, \ldots, X_n$ are independent and identically distributed random variables from a normal distribution:

$$X_i \sim \mathcal{N}(\mu, \sigma^2)$$

Find the Maximum Likelihood Estimators (MLEs) in the following cases:

(a) Find the MLE of μ when σ^2 is known.

(b) Find the MLE of σ^2 when μ is known.

(c) Find the MLEs of both μ and σ^2 simultaneously.

Solution: The likelihood function is:

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i - \mu)^2}{2\sigma^2}\right)$$

$$= \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2\right)$$

The log-likelihood is:

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2$$

Examples on MLE

(a) MLE of μ when σ^2 is known:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (X_i - \mu) = 0$$

Solving:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

(b) MLE of σ^2 when μ is known:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^{n} (X_i - \mu)^2 = 0$$

Solving:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (X_i - \mu)^2$$

Examples on MLE

(c) MLEs of both μ and σ^2 simultaneously:

To get joint ML estimators for μ and σ^2, we have to solve together the two equations

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu) = 0$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(X_i - \mu)^2 = 0$$

Solving these equations for μ and σ^2, we get

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \hat{\mu})^2$$

- MLE of μ is the sample mean.
- MLE of σ^2 is the sample variance (with divisor n, not $n-1$).

Examples on MLE

Example 3: Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables from a poisson distribution,

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

Find the MLE of λ.

Examples on MLE

Solution: The likelihood function is:

$$L(\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = e^{-n\lambda}\lambda^{\sum x_i} \cdot \prod \frac{1}{x_i!}$$

The log-likelihood function is:

$$\ell(\lambda) = -n\lambda + \left(\sum_{i=1}^{n} x_i\right) \log \lambda + \text{const}$$

Differentiate and set to zero:

$$\frac{d\ell}{d\lambda} = -n + \frac{\sum x_i}{\lambda} = 0 \Rightarrow \hat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$$

MLE of λ is the sample mean.

Examples on MLE

Example 4: Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables from an exponential distribution,

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

Find the MLE of λ.

Examples on MLE

Solution: The likelihood function is:

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum x_i}$$

The log-likelihood function is:

$$\ell(\lambda) = n \log \lambda - \lambda \sum x_i$$

Differentiate and set to zero:

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum x_i = 0 \Rightarrow \hat{\lambda} = \frac{n}{\sum x_i}$$

MLE of λ is the reciprocal of the sample mean.

Examples on MLE

Example 5: Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables from a binomial distribution,

$$f(x;p) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n$$

Find the maximum likelihood estimator of p when:

(a) a single observation is taken;

(b) a sample of m observations $X_1, X_2, \ldots, X_m$ is taken.

Solution:

i) MLE of p from a Single Observation
Let a single observation X be drawn from $\text{Bin}(n, p)$. The likelihood function is:

$$L(p) = \binom{n}{X} p^X (1-p)^{n-X}$$

Taking the natural logarithm:

$$\ln L(p) = \ln \binom{n}{X} + X \ln p + (n-X) \ln(1-p)$$

Differentiate with respect to p:

$$\frac{d}{dp} \ln L(p) = \frac{X}{p} - \frac{n-X}{1-p}$$

Set derivative to zero:

$$\frac{X}{p} = \frac{n-X}{1-p} \Rightarrow X(1-p) = (n-X)p \Rightarrow X - Xp = np - Xp \Rightarrow X = np \Rightarrow \hat{p} =$$

Examples on MLE

Answer: The MLE of p is $\boxed{\hat{p} = \frac{X}{n}}$

(ii) MLE of p from a Sample of m Observations

Let $X_1, X_2, \ldots, X_m$ be an i.i.d. sample from $\text{Bin}(n, p)$.

The likelihood function is:

$$L(p) = \prod_{i=1}^{m} \binom{n}{X_i} p^{X_i} (1-p)^{n-X_i}$$

Log-likelihood:

$$\ln L(p) = \sum_{i=1}^{m} \left[\ln \binom{n}{X_i} + X_i \ln p + (n - X_i) \ln(1-p) \right]$$

Since the binomial coefficients are constant with respect to p, we focus on:

$$\ln L(p) = \text{const} + \left(\sum_{i=1}^{m} X_i \right) \ln p + \left(mn - \sum_{i=1}^{m} X_i \right) \ln(1-p)$$

Let $T = \sum_{i=1}^{m} X_i$, then:

$$\ln L(p) = \text{const} + T \ln p + (mn - T) \ln(1-p)$$

Examples on MLE

Differentiate and set to zero:

$$\frac{d}{dp}\ln L(p) = \frac{T}{p} - \frac{mn - T}{1 - p} = 0 \Rightarrow T(1-p) = (mn-T)p \Rightarrow T = mnp \Rightarrow \hat{p} =$$

Answer: The MLE of p is $\boxed{\hat{p} = \frac{1}{mn}\sum_{i=1}^{m} X_i}$